# Chapter 3 - Differential Calculus

## 3.1 Differentiability (Differenzierbarkeit)

### ◆ Concept of Rate of Change (Änderungsrate)

We ask: *How strongly does a quantity change?*
Examples:

- Physics (Physik): displacement → velocity

$$v = \frac{\Delta s}{\Delta t}$$

- Electrical engineering (Elektrotechnik): charge → current

$$I = \frac{\Delta q}{\Delta t}$$

- Economics (Wirtschaft): price index → inflation rate

Mathematically, this is the **slope (Steigung)** of the function $f(x)$ at point $(x_0, y_0)$.

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### ◆ Definition: Derivative (Ableitung)

A function $f$ is **differentiable (differenzierbar)** at $x_0$ if the following limit exists:

$$f'(x_0) = \lim_{\Delta x \to 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}$$

- $f'(x_0)$: derivative (Ableitung) of $f$ at $x_0$
- Also written as:

$$f'(x_0) = \frac{dy}{dx}\Big|_{x_0}$$

- Represents the **instantaneous slope (Momentansteigung)** of the tangent (Tangente).

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### ◆ Basic Derivatives (Ableitungen elementarer Funktionen)

| Function $f(x)$ | Derivative $f'(x)$ | Notes |
|---|---|---|
| $x^n$ | $n \cdot x^{n-1}$ | Power rule (Potenzregel) |
| $\sin x$ | $\cos x$ | |
| $\cos x$ | $-\sin x$ | |
| $\tan x$ | $\frac{1}{\cos^2 x}$ | |

| Function $f(x)$ | Derivative $f'(x)$ | Notes |
|---|---|---|
| $e^x$ | $e^x$ | Exponential function (Exponentialfunktion) |
| $a^x$ | $(\ln a) \cdot a^x$ | |
| $\sinh x$ | $\cosh x$ | Hyperbolic sine (Sinus hyperbolicus) |
| $\cosh x$ | $\sinh x$ | Hyperbolic cosine (Kosinus hyperbolicus) |
| $\tanh x$ | $\frac{1}{\cosh^2 x}$ | Hyperbolic tangent (Tangens hyperbolicus) |

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## 3.2 Differentiation Rules (Ableitungsregeln)

### ◆ Constant Factor Rule (Faktorregel)

$$(C \cdot u(x))' = C \cdot u'(x)$$

### ◆ Sum Rule (Summenregel)

$$(f_1(x) + f_2(x) + \cdots + f_n(x))' = f_1'(x) + f_2'(x) + \cdots + f_n'(x)$$

### ◆ Product Rule (Produktregel)

$$(u(x) \cdot v(x))' = u'(x) \cdot v(x) + u(x) \cdot v'(x)$$

### ◆ Quotient Rule (Quotientenregel)

$$\left(\frac{u(x)}{v(x)}\right)' = \frac{u'(x) \cdot v(x) - u(x) \cdot v'(x)}{[v(x)]^2}$$

### ◆ Chain Rule (Kettenregel)

If $f(x) = u(v(x))$, then:

$$f'(x) = v'(x) \cdot u'(v(x))$$

Mnemonic: *inner derivative times outer derivative*
(innere Ableitung mal äußere Ableitung)

### ◆ Derivative of Inverse Function (Ableitung der Umkehrfunktion)

$$(f^{-1})'(x) = \frac{1}{f'(f^{-1}(x))}$$

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## 3.3 Curve Discussion & Extremum Problems (Kurvendiskussion und Extremwertaufgaben)

### ◆ Monotonicity (Monotonieverhalten)

- $f'(x) > 0$ → increasing (monoton steigend)
- $f'(x) < 0$ → decreasing (monoton fallend)

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## 🔹 Curvature (Krümmung)

- $f''(x_0) > 0$ → concave up (linkskrümmung)
- $f''(x_0) < 0$ → concave down (rechtskrümmung)

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## 🔹 Relative Extrema (relative Extrema)

- **Necessary condition (notwendige Bedingung):** $f'(x_0) = 0$
- **Sufficient condition (hinreichende Bedingung):**
  - $f''(x_0) > 0$ → local minimum (relatives Minimum)
  - $f''(x_0) < 0$ → local maximum (relatives Maximum)
- $f'(x_0) = 0$, $f''(x_0) = 0$ → possible saddle point (Sattelstelle)

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## 🔹 Inflection Points (Wendepunkte)

- $f''(x_0) = 0$ and $f'''(x_0) \neq 0$
- If also $f'(x_0) = 0$ → saddle point (Sattelstelle)

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## 🔹 Complete Curve Discussion (vollständige Kurvendiskussion)

1. Domain (Definitionsbereich)
2. Range (Wertebereich)
3. Zeros (Nullstellen)
4. Poles (Polstellen)
5. Limits as $x \to \pm\infty$
6. Asymptotes (Asymptoten)
7. Extrema (Extrema)
8. Inflection/Saddle points (Wendepunkte / Sattelpunkte)
9. Graph sketch (Funktionsgraph)

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# 3.4 L'Hospital's Rule (L'Hospitalsche Regel)

Used for indeterminate forms (unbestimmte Ausdrücke).

If

$$\lim_{x \to a} f(x) = \lim_{x \to a} g(x) = 0 \quad (\text{type } 0/0)$$

or

$$\lim_{x \to a} f(x) = \lim_{x \to a} g(x) = \pm\infty \quad (\text{type } \infty/\infty)$$

then

$$\lim_{x \to a} \frac{f(x)}{g(x)} = \lim_{x \to a} \frac{f'(x)}{g'(x)}$$

Also applicable to $0 \cdot \infty$, $\infty - \infty$, $0^0$, $\infty^0$, $1^\infty$ after suitable algebraic manipulation.

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## 3.5 Newton's Method (Newton-Verfahren)

Used to find zeros (Nullstellen) of $f(x)$ approximately.

### Formula:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

- Start with an initial guess $x_0$
- Converges **quadratically (quadratische Konvergenz)** if convergence occurs

### Stopping Criteria (Abbruchkriterien)

1. $|f(x_n)| < \varepsilon$
2. $\frac{|x_{n+1} - x_n|}{|x_n|} < \delta$
3. $n = N_{max}$

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## ✅ Summary Table (Überblick)

| Concept | German Term | Formula / Note |
|---|---|---|
| Derivative | Ableitung | $f'(x) = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x}$ |
| Differentiable | differenzierbar | Limit must exist |
| Product Rule | Produktregel | $u'v + uv'$ |
| Quotient Rule | Quotientenregel | $\frac{u'v - uv'}{v^2}$ |
| Chain Rule | Kettenregel | inner × outer derivative |
| Extremum | Extremum | $f'(x) = 0, f''(x) \neq 0$ |
| Inflection Point | Wendepunkt | $f''(x) = 0, f'''(x) \neq 0$ |
| L'Hospital's Rule | Regel von L'Hospital | $\frac{f'(x)}{g'(x)}$ |
| Newton's Method | Newton-Verfahren | $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$ |